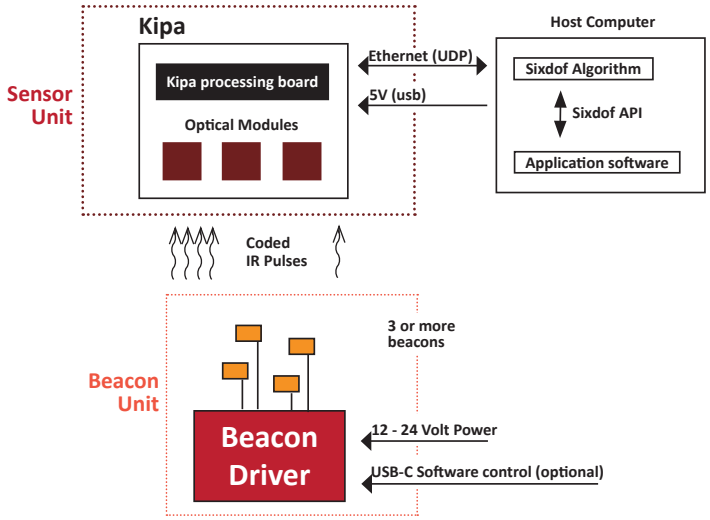
Sixdof Space
ULTRA FAST OPTICAL TRACKING

Beacon Driver Boards

Multiple IR beacons provide spatial reference for the Sixdof tracking system. Each beacon is identified by a unique ID, transmitted optically at high speed. One or more beacons are attached to a beacon driver to receive power and their unique ID. Sixdof offers multiple types of beacon drivers to enable a wide variety of use cases, from indoor use cases with restricted power consumption and payload, to outdoor use cases at far distances. Some beacon driver boards are runtime programmable.

Sixdof System Architecture

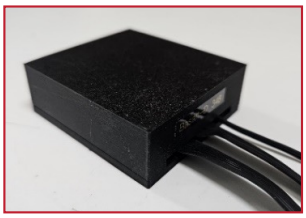


Below is a comparison of the different beacon driver boards available:

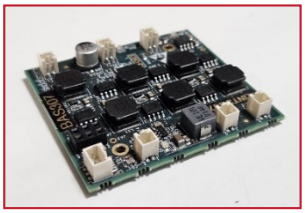
Type	Max # Beacons	Input Power	Drive Current	Runtime Programable	Dimensions	Weight	Compatible with
Basalt L	6	12V / 100mA 24V / 50mA	0.3 Amps	No	55x50x10 mm	22g	Low-Power Beacons
Basalt D	6	12V / 400mA 24V / 200mA	2 Amps	No	55x50x10 mm	22g	Low-Power Beacons
Shale	8	24V / 3A	2A, 4A, 6A*	Yes	170x60x14 mm	76g	HYF-N, HYF-S, 100W
Zircon	3	24V / 4A	up to 24 Amps*	Yes	90x60x20 mm	60g	HYF-120S-4, VCSEL

** Shale and Zircon are configurable at manufacturing time*

BASALT



Housing
Size: 70x56x24 mm
Weight: 60g



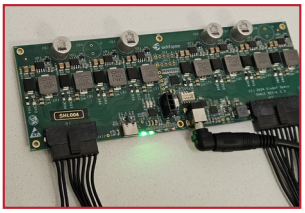
Driver Board
Size: 55x50x10 mm
Weight: 22g

020-1001-001 Basalt-L with housing
020-1002-001 Basalt-D with housing

SHALE



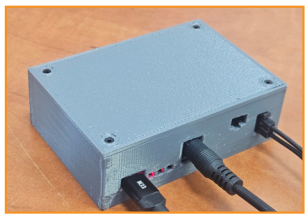
Housing
Size: 178x68x29 mm
Weight: 133g



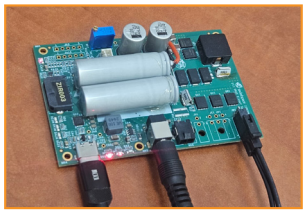
Driver Board
Size: 170x60x14 mm
Weight: 76g

020-1006-000 Shale with housing (Standard Configuration 2A/6A)
020-1006-001 Shale with housing (Custom Configuration 2A/4A/6A)

ZIRCON



Housing
Size: 95x65x30 mm
Weight: 120g



Driver Board
Size: 90x60x20 mm
Weight: 60g

020-1011-000 Zircon with housing (Custom Configuration up to 24A)

- Sixdof's beacon driver boards are compatible with multiple types of IR beacons.
- The different types of beacons vary in their minimum and maximum ranges, field of view, and size.
- All beacons operate in the 850nm wavelength, and have been tested in full sunlight.

Before purchasing a beacon driver and beacons, consult with the Sixdof team to identify the combination that best fits your use case.

Software Control Options

SIXDOF SPIDER

Sixdof Spider is the software that controls our smart family of beacon driver boards. This includes the Sixdof Zircon and Shale boards.

The Sixdof Spider SDK enables the runtime control with the following features:

- Toggle individual beacons ON and OFF at runtime
- Configure beacons with custom, unique IDs
- Send short, 8-character optical "Dynamic Messages" from the beacons to the sensor

Sixdof Spider SDK comes with full documentation, standalone examples, GUI tools, and runtime libraries, available on the following platforms:

- Windows
- Linux Ubuntu
- Linux Raspberry Pi OS (32 bit)
- Linux Raspberry Pi OS (64 bit)

Sixdof Space
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Elevate your robotics and automation capabilities with Sixdof's unmatched optical tracking precision—accelerate your operations today.

For more details, visit us at www.sixdofspace.com